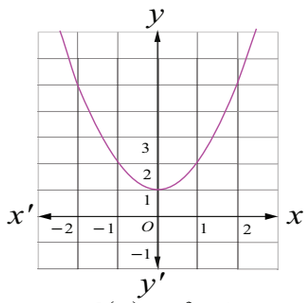


$f(x) = x^2 - 1$

Fig. 2.1

Equation 1 has two real solutions, $x = -1$ and $x = 1$. We know that solving an equation in x is equivalent to finding the x-intercepts of a graph of $f(x) = x^2 - 1$ crosses the x-axis at $(-1, 0)$ and $(1, 0)$.



$f(x) = x^2 + 1$

Fig. 2.2

By the same logic, **equation 2** has no real solutions since the graph of $f(x) = x^2 + 1$ does not cross the x-axis; we can see this by looking at the graph of $f(x) = x^2 + 1$.

This is because, when we square a real number it is impossible to get a negative real number. If equation 2 has solutions, then we must create an imaginary number as a square root of -1. This imaginary unit $\sqrt{-1}$ is denoted by i. The imaginary number i tells us that $i^2 = -1$. We can use this fact to find other powers of i.

2.1.1 Powers of imaginary unit i

$i^0 = 1,\ i^1 = i$	$i^2 = -1$	$i^3 = i^2 i = -i$	$i^4 = i^2 i^2 = 1$
$(i)^{-1} = \frac{1}{i} = \frac{i}{(i)^2} = -i$	$(i)^{-2} = -1$	$(i)^{-3} = i$	$(i)^{-4} = 1 = i^4$

We note that, for any integer n, i^n has only four possible values: they correspond to values of n when divided by 4 leave the remainders 0, 1, 2, and 3.That is when the integer $n \le -4$ or $n \ge 4$, using division algorithm, n can be written as $n = 4q + k,\ \ 0 \le k < 4,\ \ k$ and q are integers and we write

$$(i)^n = (i)^{4q+k} = (i)^{4q}(i)^k = \left((i)^4\right)^q (i)^k = (1)^q (i)^k = (i)^k$$

Example 2.1

Simplify the following

(i) i^7 (ii) i^{1729} (iii) $i^{-1924} + i^{2018}$ (iv) $\sum_{n=1}^{102} i^n$ (v) $i\, i^2\, i^3 \cdots i^{40}$

Solution

(i) $(i)^7 = (i)^{4+3} = (i)^3 = -i$; (ii) $i^{1729} = i^{1728}\, i^1 = i$

(iii) $(i)^{-1924} + (i)^{2018} = (i)^{-1924+0} + (i)^{2016+2} = (i)^0 + (i)^2 = 1 - 1 = 0$

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